# Partial Derivatives and the Gradient

**ANSWER KEY**

## Section 1: First-Order Partial Derivatives

**1** Find $f_x$ and $f_y$ for $f(x, y) = 3x^2y - 5xy^3 + 4x - 2y$.

**a.** $f_x = 6xy - 5y^3 + 4$, $f_y = 3x^2 - 15xy^2 - 2$ ✓

**b.** $f_x = 6xy - 5y^3$, $f_y = 3x^2 - 15xy^2$

**c.** $f_x = 6x - 5y^3 + 4$, $f_y = 3x^2 - 15y^2 - 2$

**d.** $f_x = 3x^2 - 15xy^2$, $f_y = 6xy - 5y^3$

Treat $y$ as constant: $f_x = 6xy - 5y^3 + 4$. Treat $x$ as constant: $f_y = 3x^2 - 15xy^2 - 2$.

**2** Find $\frac{\partial z}{\partial x}$ for $z = e^{x^2y}$.

**a.** $2xye^{x^2y}$ ✓

**b.** $x^2e^{x^2y}$

**c.** $2xe^{x^2y}$

**d.** $2xye^{2xy}$

By the chain rule: $\frac{\partial z}{\partial x} = e^{x^2y}\frac{\partial}{\partial x}(x^2y) = 2xye^{x^2y}$.

**3** Find $f_y(1, 0)$ for $f(x, y) = \ln(x^2 + y^2 + 1)$.

**a.** 0 ✓

**b.** 1

**c.** 2

**d.** 1/2

$f_y = \frac{2y}{x^2 + y^2 + 1}$. Evaluating at $(1, 0)$: $f_y(1, 0) = \frac{0}{1 + 0 + 1} = 0$.

---

## Section 2: Second-Order Partials and Clairaut's Theorem

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

**4** What does Clairaut's Theorem (equality of mixed partials) state for continuous second partial derivatives?

**a.** $f_{xy} = f_{yx}$ ✓

**b.** $f_{xx} = f_{yy}$

**c.** $f_{xy} = -f_{yx}$

**d.** $f_{xx} + f_{yy} = 0$

If the mixed partial derivatives $f_{xy}$ and $f_{yx}$ are both continuous on an open set, they are equal: $f_{xy} = f_{yx}$.

**5** Find $f_{xx}$ for $f(x, y) = x^4y^2 - 2x^2y$.

**a.** $12x^2y^2 - 4y$ ✓

**b.** $4x^3y^2 - 4xy$

**c.** $2x^4 - 0$

**d.** $12x^2 - 4$

$f_x = 4x^3y^2 - 4xy$. Differentiating with respect to $x$ again: $f_{xx} = 12x^2y^2 - 4y$.

**6** Verify whether $u(x, y) = e^x \sin y$ satisfies Laplace's equation $u_{xx} + u_{yy} = 0$.

**a.** **Yes,** $u_{xx} + u_{yy} = e^x \sin y - e^x \sin y = 0$ ✓

**b.** No, $u_{xx} + u_{yy} = 2e^x \sin y$

**c.** No, it equals $e^x \cos y$

**d.** Only when $x = 0$

$u_x = e^x \sin y \Rightarrow u_{xx} = e^x \sin y$. $u_y = e^x \cos y \Rightarrow u_{yy} = -e^x \sin y$. $u_{xx} + u_{yy} = 0$.

---

## Section 3: The Gradient Vector and Directional Derivatives

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

**7** Find the gradient $\nabla f(x, y)$ for $f(x, y) = x^2y + 3y^2$ at the point $(2, 1)$.

**a.** $\langle 4, 10 \rangle$ ✓

**b.** $\langle 4, 7 \rangle$

**c.** $\langle 2, 6 \rangle$

**d.** $\langle 10, 4 \rangle$

$\nabla f = \langle 2xy, x^2 + 6y \rangle$. At $(2, 1)$: $\nabla f(2, 1) = \langle 2(2)(1), 4 + 6(1) \rangle = \langle 4, 10 \rangle$.

**8** In which direction does a differentiable function $f$ increase most rapidly at a point $P$?

**a.** **In the direction of the gradient vector** $\nabla f(P)$ ✓

**b.** Opposite to $\nabla f(P)$

**c.** Orthogonal to $\nabla f(P)$

**d.** Along the level curve

The directional derivative $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u} = \|\nabla f\| \cos\theta$ is maximized when $\cos\theta = 1$, which is the direction of $\nabla f$.

**9** Find the directional derivative of $f(x, y) = 2x^2 - 3y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 3, 4 \rangle$.

**a.** $-12/5$ ✓

**b.** $12/5$

**c.** $-24$

**d.** $0$

$\nabla f = \langle 4x, -6y \rangle \Rightarrow \nabla f(1, 1) = \langle 4, -6 \rangle$. Unit vector $\mathbf{u} = \langle 3/5, 4/5 \rangle$.
$D_{\mathbf{u}} f = \langle 4, -6 \rangle \cdot \langle 3/5, 4/5 \rangle = 12/5 - 24/5 = -12/5$.

---

## Section 4: Tangent Planes and Linear Approximations

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

**10** Find the equation of the tangent plane to $z = x^2 + y^2$ at the point $(1, 2, 5)$.

**a.** $z - 5 = 2(x - 1) + 4(y - 2)$ ✓

**b.** $z - 5 = (x - 1) + 2(y - 2)$

**c.** $2x + 4y - z = 0$

**d.** $z = 2x + 4y + 5$

$z_x = 2x \Rightarrow z_x(1, 2) = 2$. $z_y = 2y \Rightarrow z_y(1, 2) = 4$. Plane: $z - 5 = 2(x - 1) + 4(y - 2) \Rightarrow 2x + 4y - z = 5$.

**11** Use the linear approximation $L(x, y)$ of $f(x, y) = \sqrt{x^2 + y^2}$ at $(3, 4)$ to estimate $f(3.02, 3.97)$.

**a.** $4.988$ ✓

**b.** $5.012$

**c.** $5.000$

**d.** $4.950$

$f(3, 4) = 5$. $f_x = x/\sqrt{x^2 + y^2} = 3/5 = 0.6$; $f_y = 4/5 = 0.8$.
$L = 5 + 0.6(0.02) + 0.8(-0.03) = 5 + 0.012 - 0.024 = 4.988$.

**12** Find the normal vector to the level surface $x^2 + y^2 + z^2 = 9$ at $(1, 2, 2)$.

**a.** $\langle 2, 4, 4 \rangle$

**b.** $\langle 1, 2, 2 \rangle$

**c.** **Both** $\langle 1, 2, 2 \rangle$ **and** $\langle 2, 4, 4 \rangle$ **are valid normal vectors** ✓

**d.** $\langle 2, 2, 1 \rangle$

The gradient of $F(x, y, z) = x^2 + y^2 + z^2$ is $\nabla F = \langle 2x, 2y, 2z \rangle = \langle 2, 4, 4 \rangle = 2\langle 1, 2, 2 \rangle$, which is normal to the tangent plane.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647